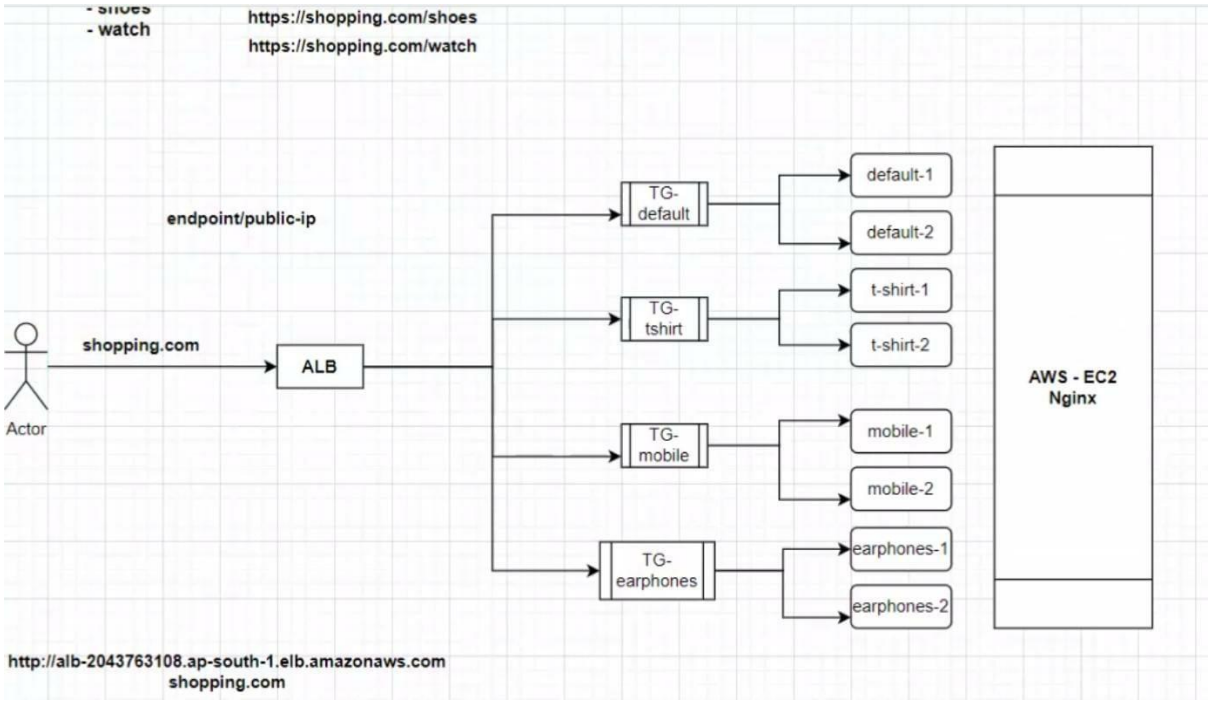
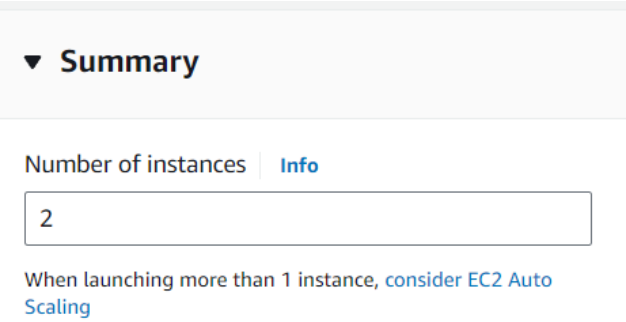
Application Load Balancer (ALB)

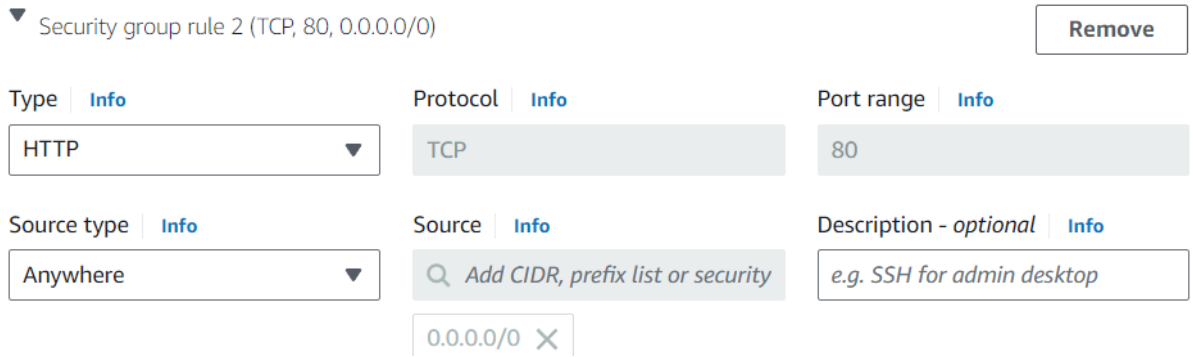


Create 8 instances as per diagram

When instance creation, enter number of instances to be created. (I.e. for default enter 2)

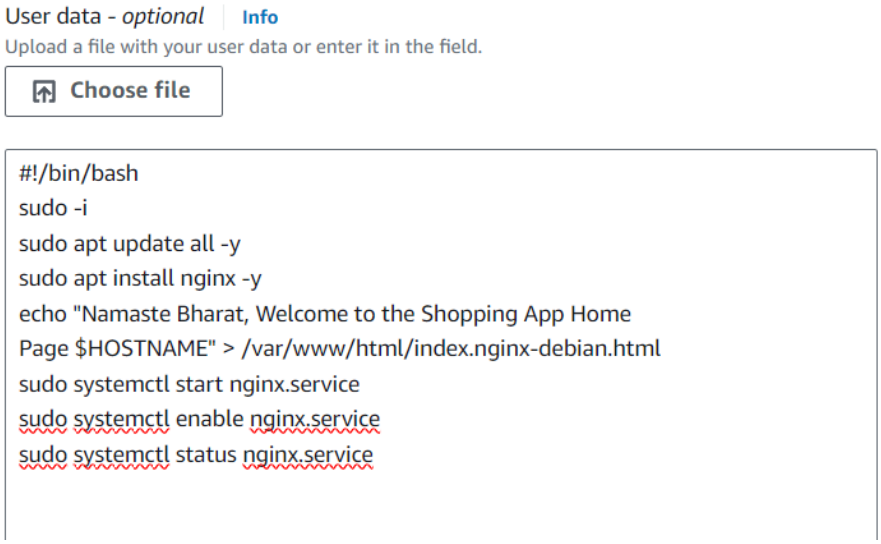


Edit Network Setting - Change security group to-

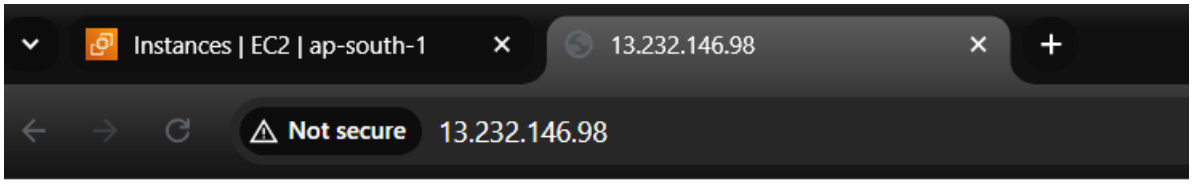


Advance Details -> Add user data (Script)

```
#!/bin/bash
sudo -i
sudo apt update all -y
sudo apt install nginx -y
echo "Namaste Bharat, Welcome to the Shopping App Home
Page $HOSTNAME" > /var/www/html/index.nginx-debian.html
sudo systemctl start nginx.service
sudo systemctl enable nginx.service
sudo systemctl status nginx.service
```



Copy public IP of both instances & check it on URL that works or not.



Namaste Bharat, Welcome to the Shopping App Home Page ip-172-31-47-105

Create remaining T-Shirt, Mobile & Earphone instances (2 of each)

Only change is the **User Data (Script).**

T-shirt----------

```
#!/bin/bash
sudo -i
sudo apt update all -y
sudo apt install nginx -y
mkdir -p /var/www/html/tshirt/
echo "hello world, welcome to my tshirt web-page
$HOSTNAME" > /var/www/html/tshirt/index.html
sudo systemctl start nginx.service
sudo systemctl enable nginx.service
sudo systemctl status nginx.service
```

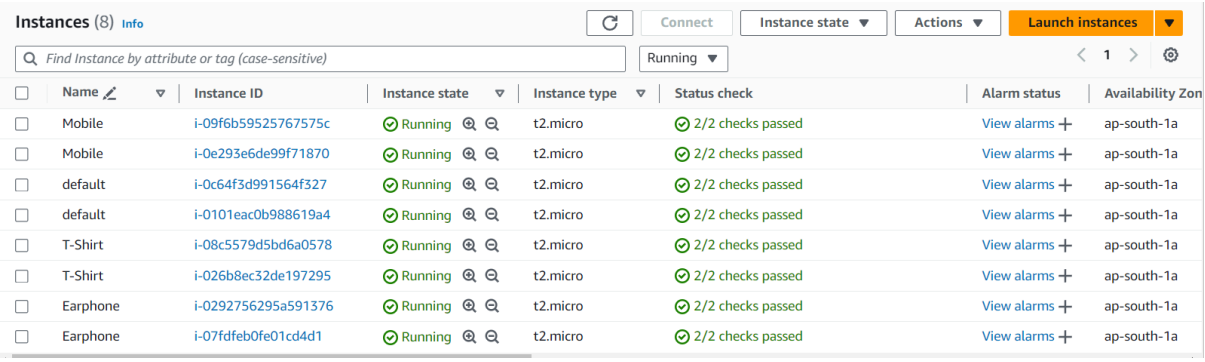
Mobile----------

```
#!/bin/bash
sudo-i
sudo apt update all -y
sudo apt install nginx -y
mkdir -p /var/www/html/mobile/
echo "hello world, welcome to my mobile web-page
$HOSTNAME" > /var/www/html/mobile/index.html
sudo systemctl start nginx.service
sudo systemctl enable nginx.service
sudo systemctl status nginx.service
```

Earphone----------

```
#!/bin/bash
sudo -i
sudo apt update all -y
sudo apt install nginx -y
mkdir -p /var/www/html/earphones/
echo "hello world, welcome to my earphones web-page
$HOSTNAME" > /var/www/html/earphones/index.html
sudo systemctl start nginx.service
sudo systemctl enable nginx.service
sudo systemctl status nginx.service
```

Instances looks like this



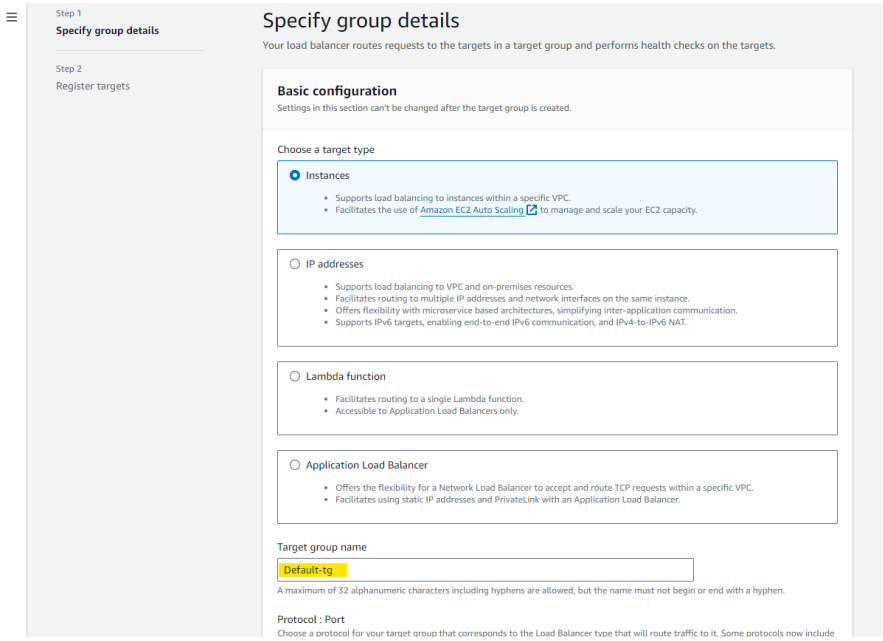
Instances (8) Info

Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | Running | 1

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zon
Mobile	i-09f6b59525767575c	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a
Mobile	i-0e293e6de99f71870	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a
default	i-0c64f3d991564f327	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a
default	i-0101eac0b988619a4	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a
T-Shirt	i-08c5579d5bd6a0578	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a
T-Shirt	i-026b8ec32de197295	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a
Earphone	i-0292756295a591376	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a
Earphone	i-07fdfeb0fe01cd4d1	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a

Target Groups Creation- Add Target Group Name only & → Next



Step 1
Specify group details

Step 2
Register targets

Specify group details

Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

Instances
- Supports load balancing to instances within a specific VPC.
- Facilitates the use of Amazon EC2 Auto Scaling to manage and scale your EC2 capacity.

IP addresses
- Supports load balancing to VPC and on-premises resources.
- Facilitates routing to multiple IP addresses and network interfaces on the same instance.
- Offers flexibility with microservice based architectures, simplifying inter-application communication.
- Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.

Lambda function
- Facilitates routing to a single Lambda function.
- Accessible to Application Load Balancers only.

Application Load Balancer
- Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
- Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name

Default-tg

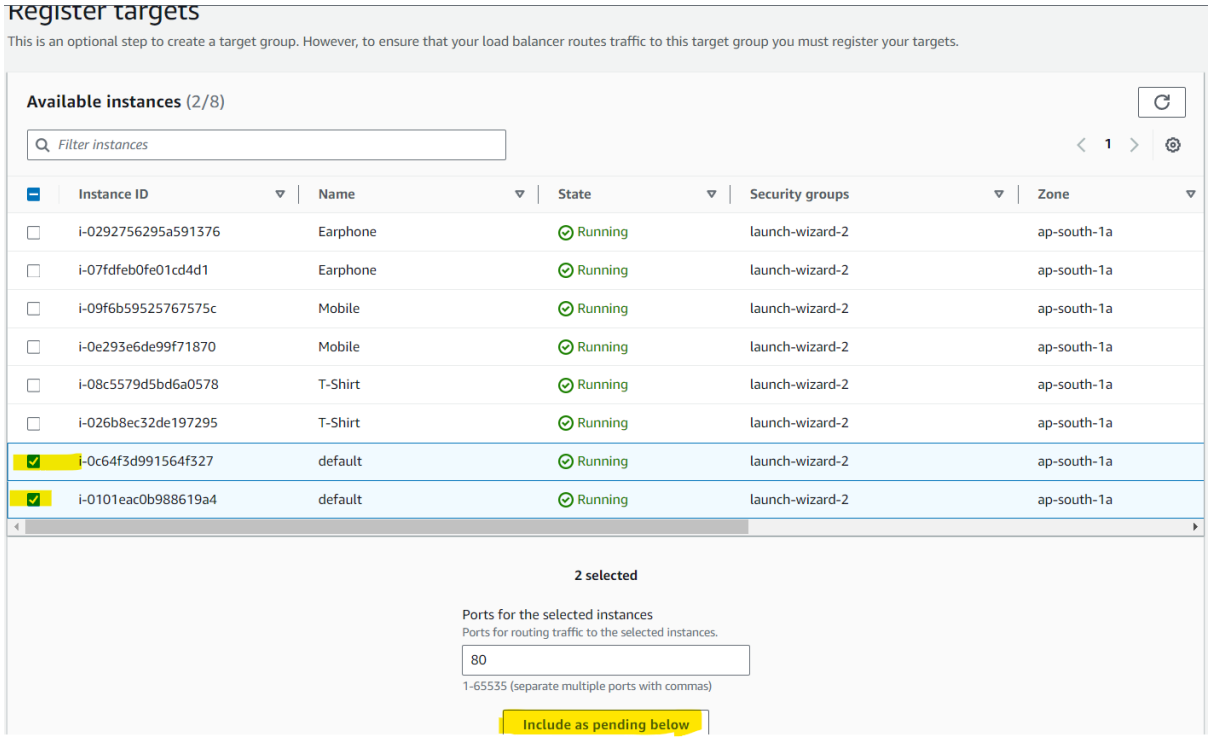
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol : Port

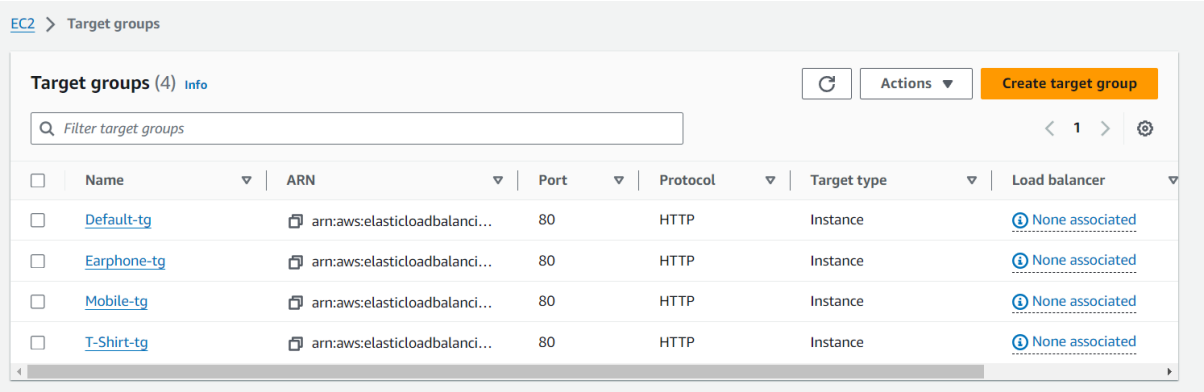
Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include

Register Targets-

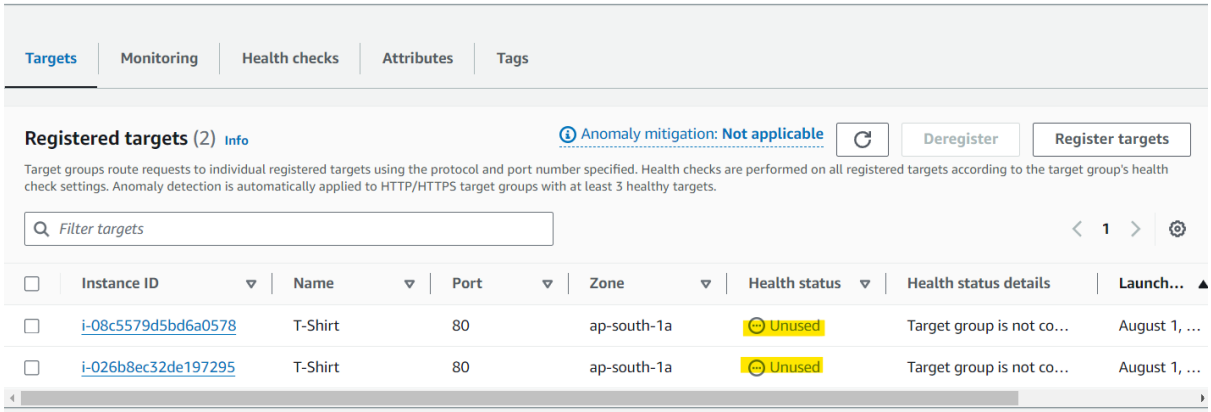
Select same two instance target & Include as pending below→ Create Target Group



Create same for remaining



Here Health Status shows to 'Unused'



Load Balancer Create- Select Application Load Balancer(ALB)-

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

ALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme Info

Scheme can't be changed after the load balancer is created.

- (●) Internet-facing
  An internet-facing load balancer routes requests from clients over the internet to targets. Requires a public subnet. Learn more
- ( ) Internal
  An internal load balancer routes requests from clients to targets using private IP addresses. Compatible with the **IPv4** and **Dualstack** IP address types.

Select all Availability Zones

Availability Zones

[x] **ap-south-1a (aps1-az1)**

Subnet

subnet-08bcd76d7fdc29343
IPv4 subnet CIDR: 172.31.32.0/20

IPv4 address
Assigned by AWS

[x] **ap-south-1b (aps1-az3)**

Subnet

subnet-0f68909eb75b26a98
IPv4 subnet CIDR: 172.31.0.0/20

IPv4 address
Assigned by AWS

[x] **ap-south-1c (aps1-az2)**

Subnet

subnet-0e2a15341909605d9
IPv4 subnet CIDR: 172.31.16.0/20

IPv4 address
Assigned by AWS

Select your Security Group

Security groups Info

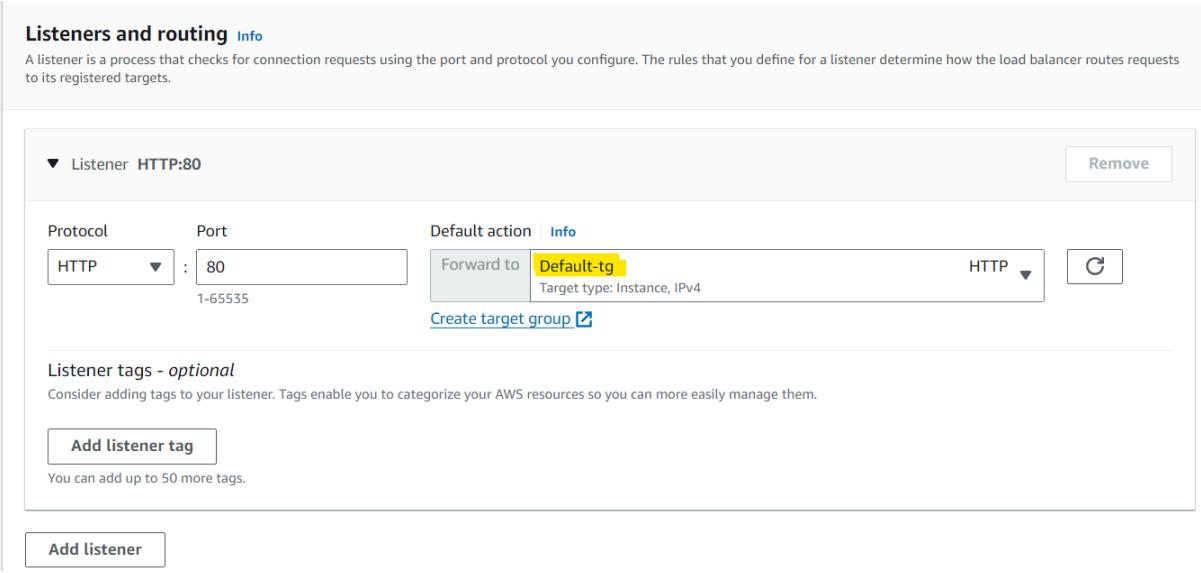
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can create a new security group.

Security groups

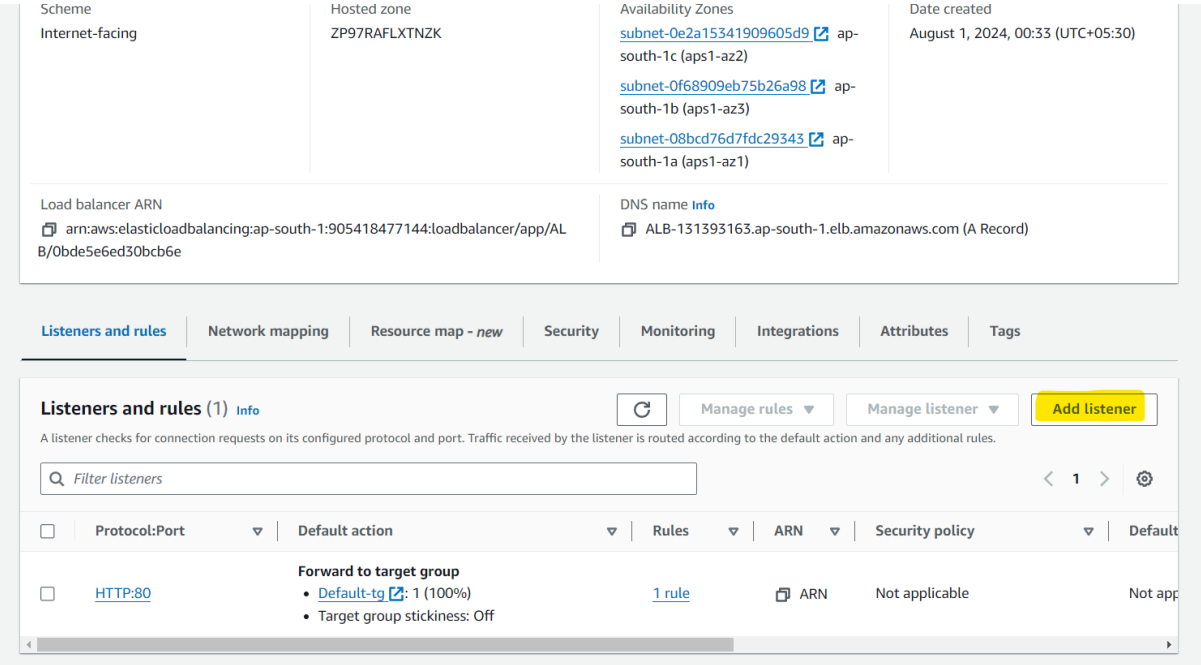
Select up to 5 security groups

launch-wizard-2 ×
sg-00f3ea59aa1c4b0b8 VPC: vpc-04fda53d2f783b4c2

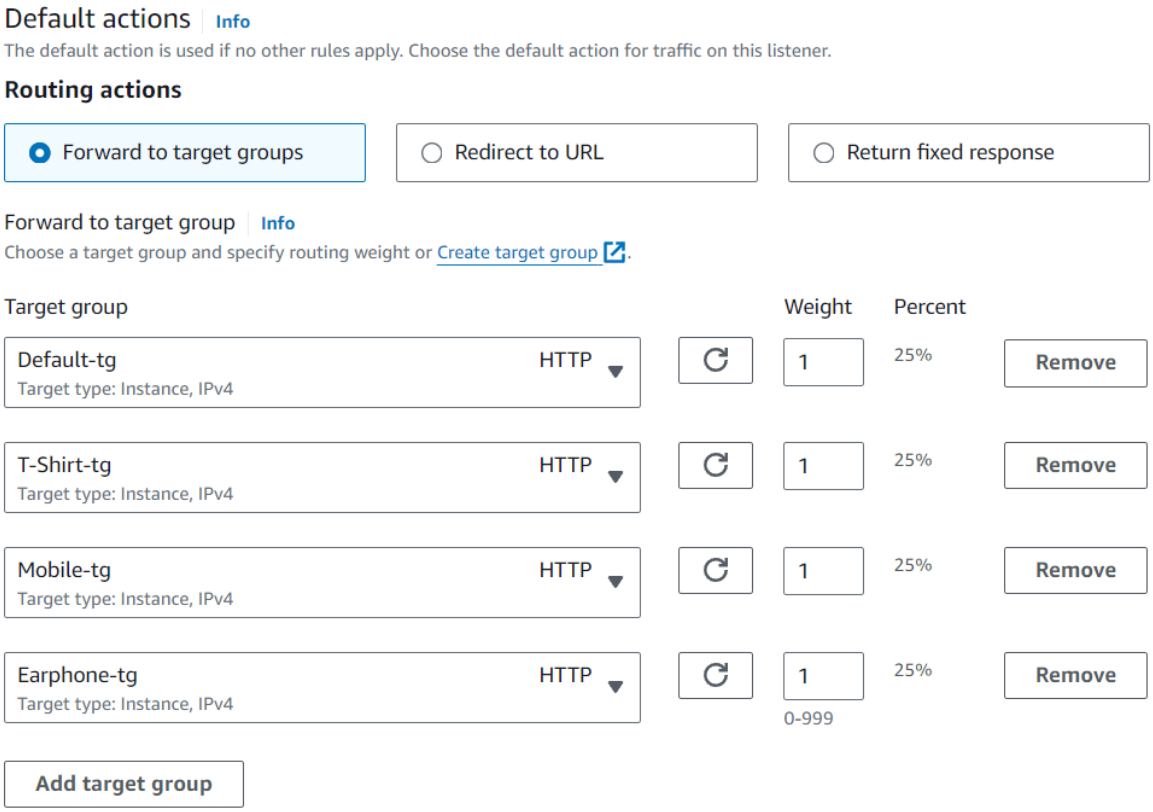
Listeners & Rounting→ select Default-tg (Target group) as default action→Create Load Balancer



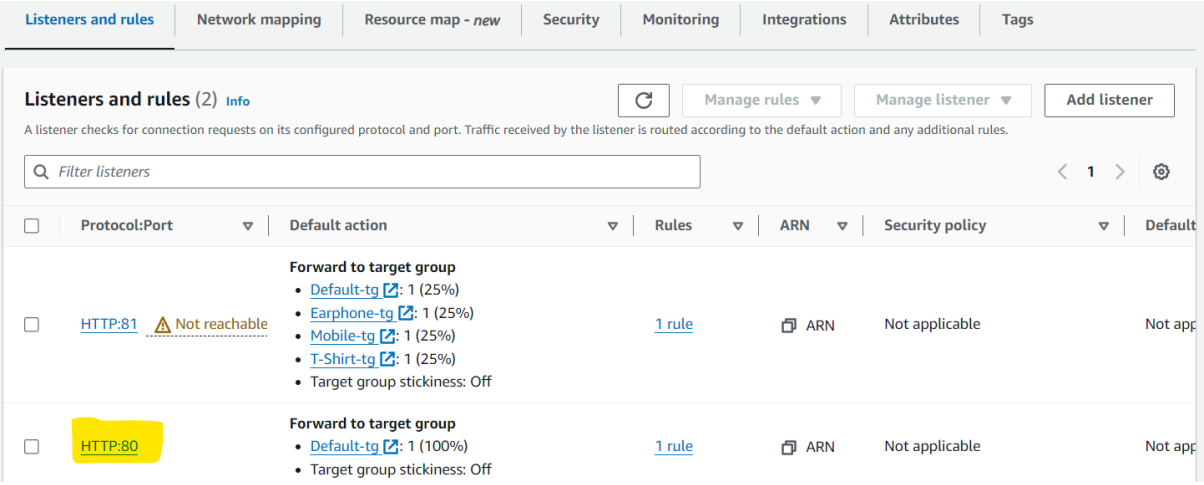
Listeners & Rules→ Add Listener



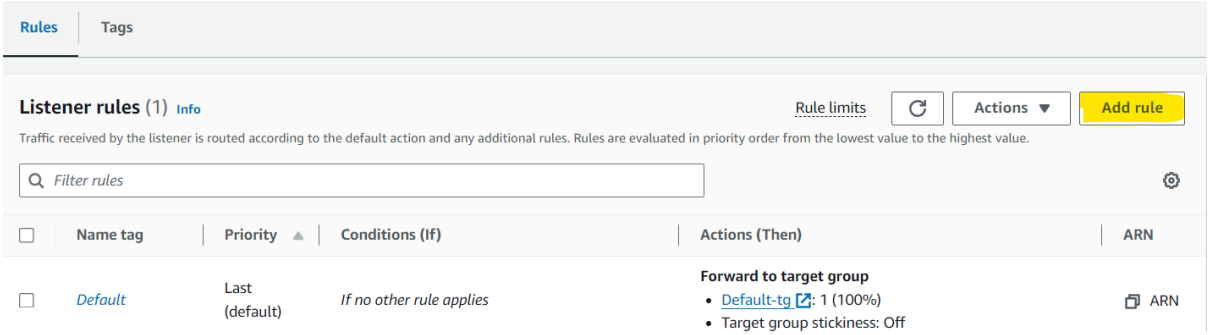
Select & Set Target Groups→ Add



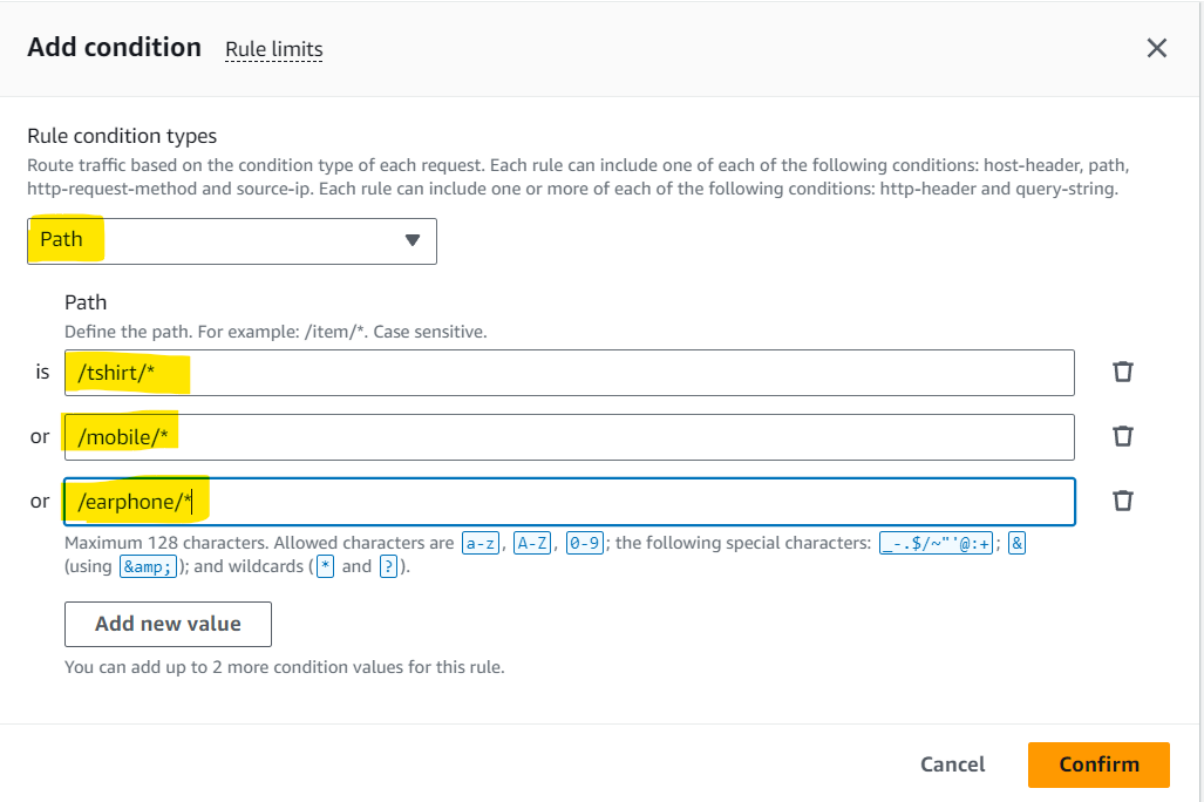
In we created Application Load Balancer→ Listeners & Rules→ HTTP:80

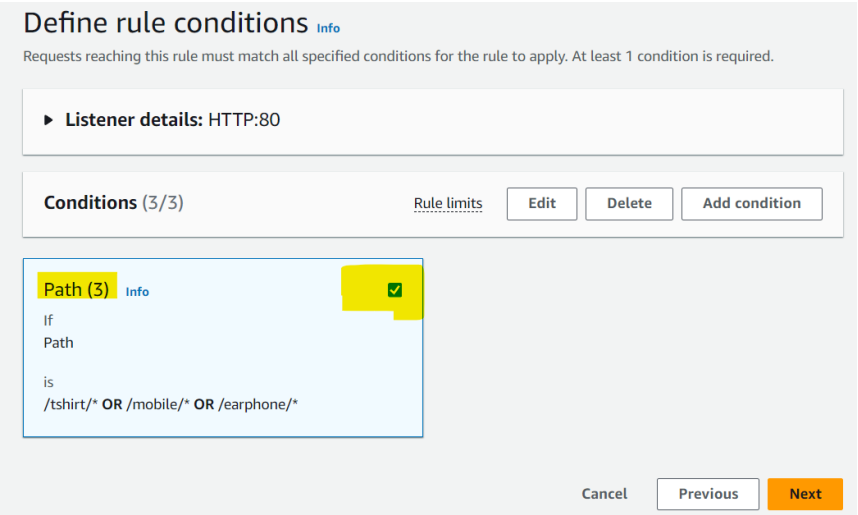


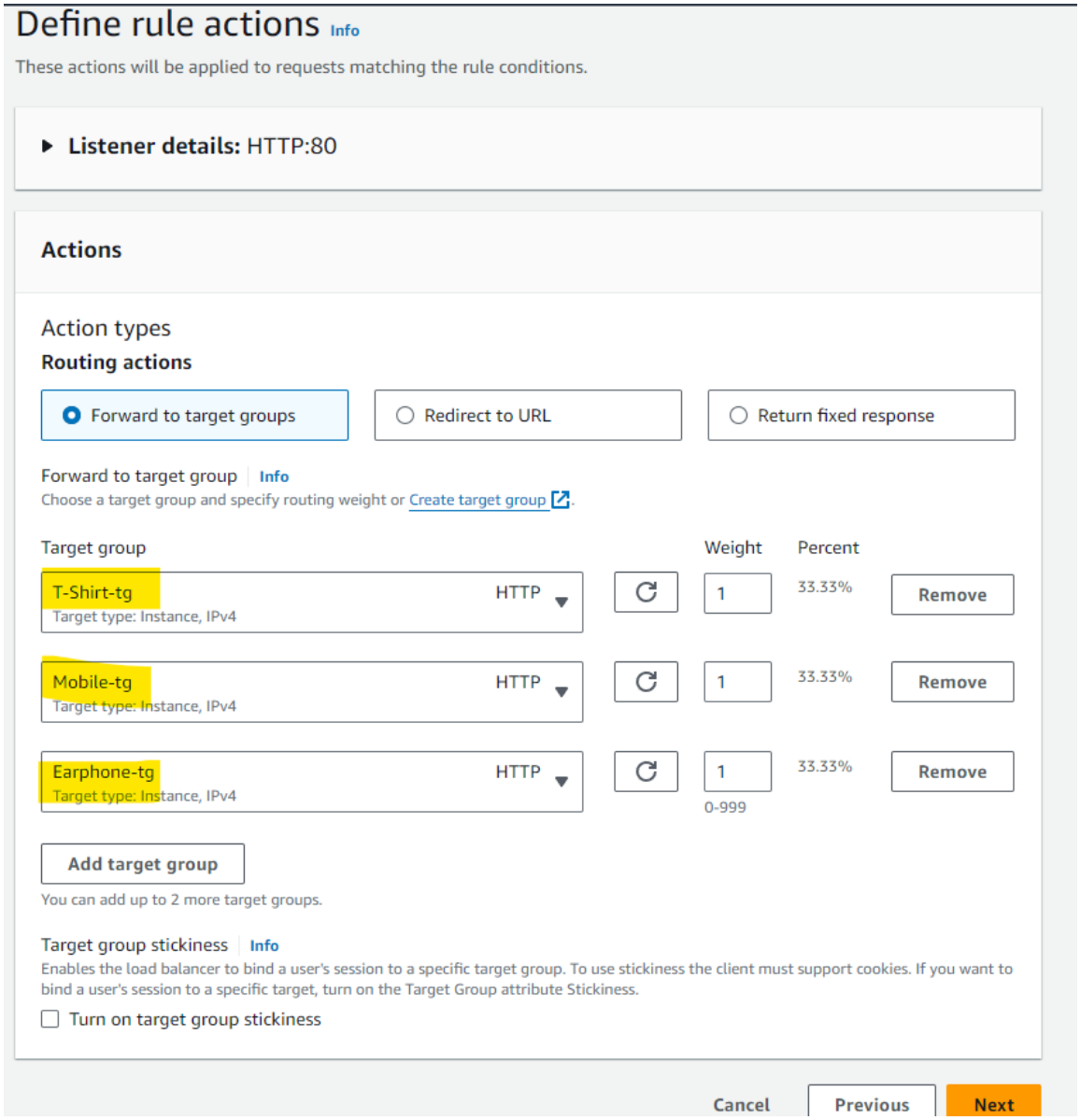
→Add Rule→ Name it (My Rules)



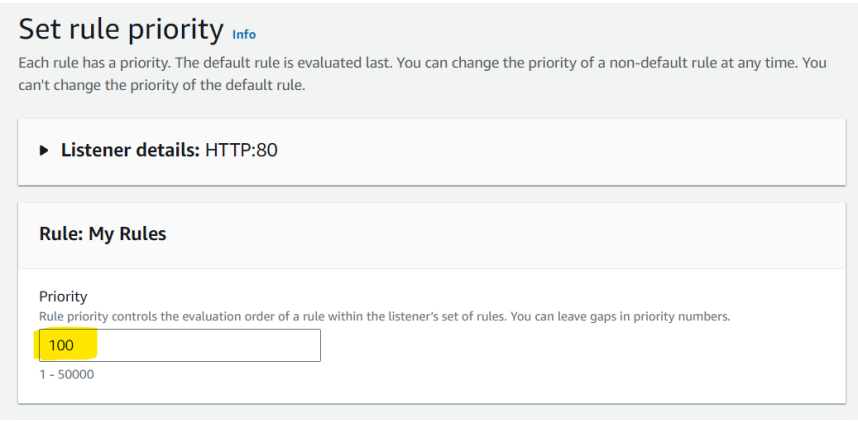
Add Condition→ select Path→ define path as per named in Script→Next



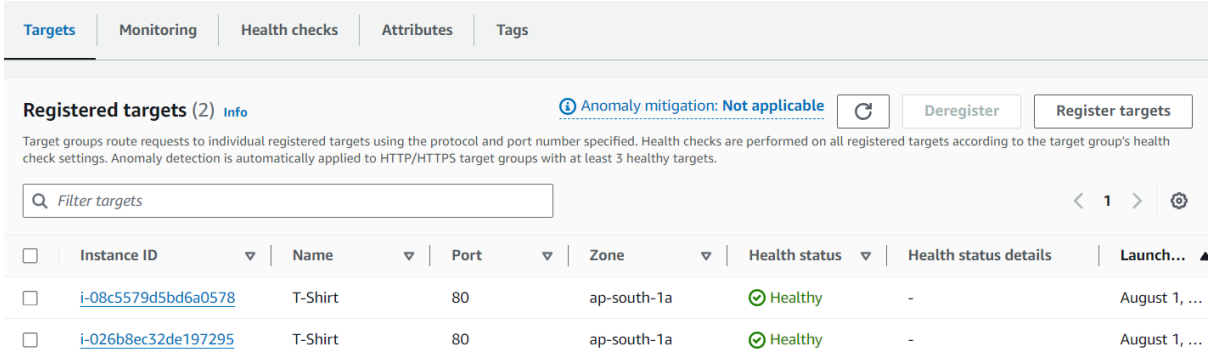




Set Rule Priority→Next→ Create



Check in to Target Group Health Status→ It changes **Unused** to **Healthy** for all instances.



Finally,

Copy DNS from instance & paste it to browser with (DNS/tshirt) (DNS/mobile) (DNS/earphones)

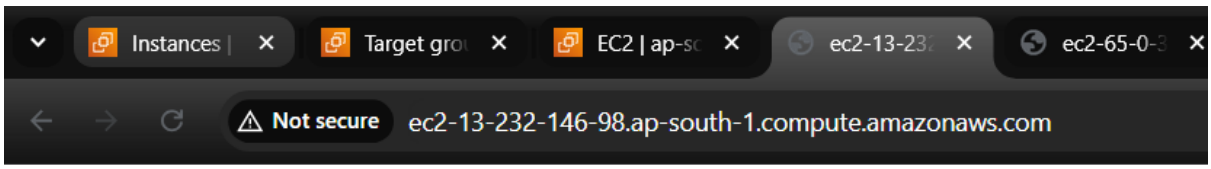
Example-

http://ec2-65-0-31-5.ap-south-1.compute.amazonaws.com/tshirt/

http://ec2-15-206-67-61.ap-south-1.compute.amazonaws.com/mobile/

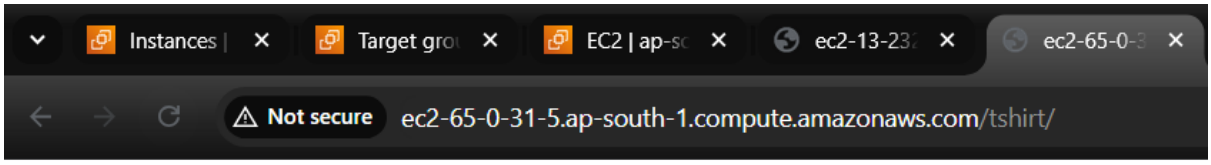
http://ec2-13-126-159-20.ap-south-1.compute.amazonaws.com/earphones/

Default instance- Only with DNS



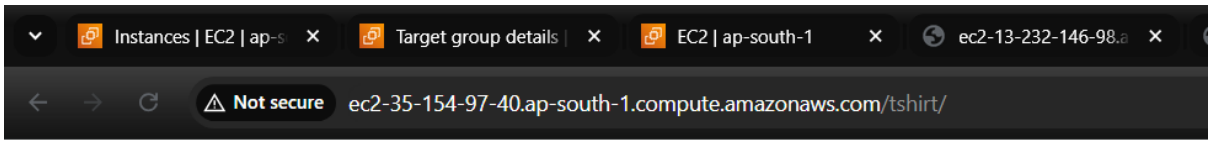
Namaste Bharat, Welcome to the Shopping App Home Page ip-172-31-47-105

T-Shirt 1



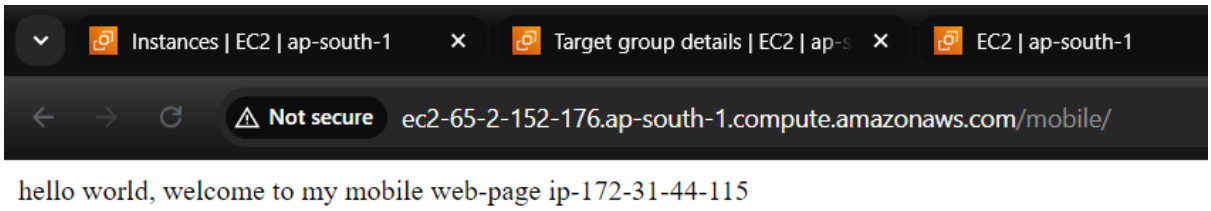
hello world, welcome to my tshirt web-page ip-172-31-44-217

T-Shirt 2

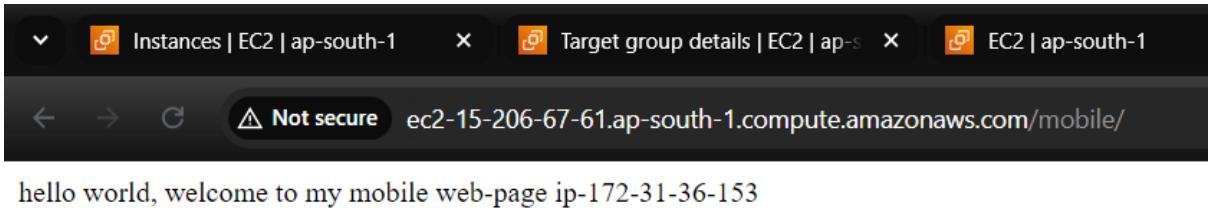


hello world, welcome to my tshirt web-page ip-172-31-43-57

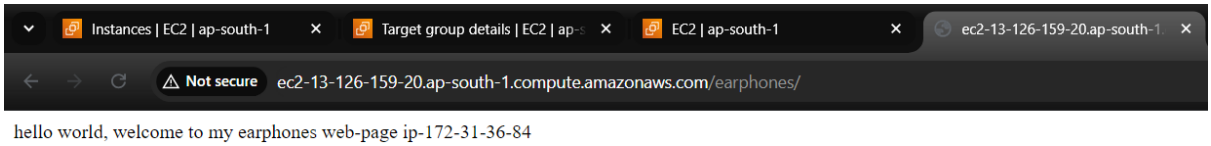
Mobile 1



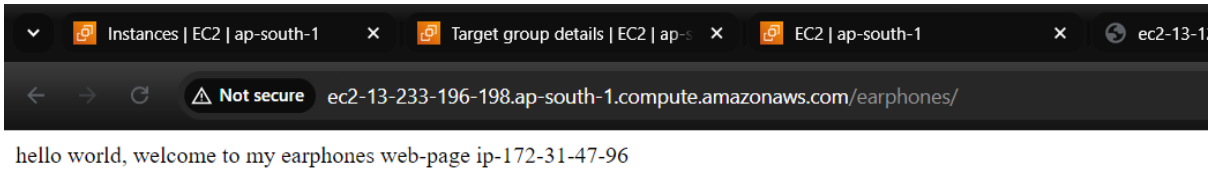
Mobile 2

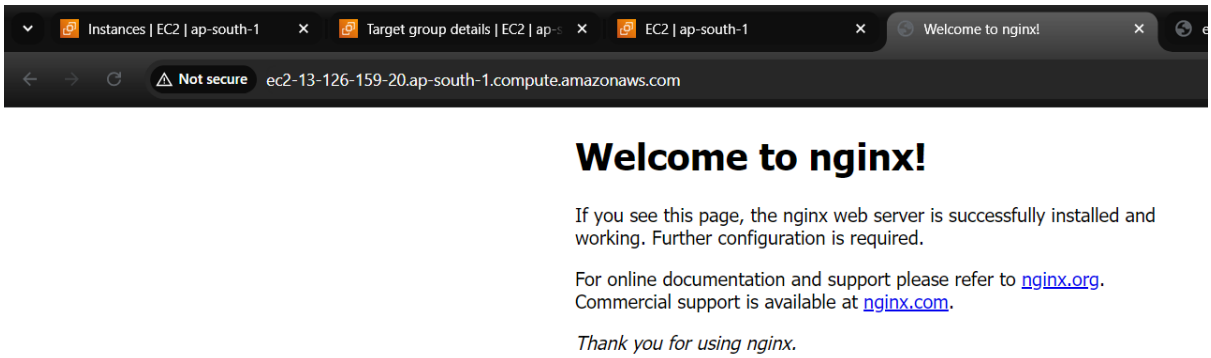


Earphone 1



Earphone 2



.

It happens when only paste DNS (not DNS/tshirt or DNS/mobile)